

SEDS SOURCING BRIDGE

Web Portal & Engineering Intake Architecture Specification

System Implementation Guide for Frontend / Full-Stack Webmaster

Project Sponsor: SEDS Pakistan · Space Technology & Robotics Pipeline

Project Lead: Muhammad Zubair Mongol (President, SEDS Pakistan | SJTU Fellow)

Operational Base: Shanghai Jiao Tong University / Yangtze River Delta Precision Corridor

Classification: Internal Engineering Technical Brief | **Status:** Active Staging Brief

1 Executive Summary & Context

The **SEDS Sourcing Bridge** is a dedicated digital and operational infrastructure connecting collegiate rocketry teams, CubeSat developers, and university hardware innovators worldwide with verified precision manufacturing bases across China's Yangtze River Delta (Shanghai/Kunshan/Suzhou) and Greater Bay Area (Shenzhen/Zhuhai).

1.1 The Problem Being Solved

Overseas student aerospace teams and hardware startups face crippling local manufacturing bottlenecks:

- **Excessive Low-Volume Markups:** Traditional Western shops charge \$3,000–\$5,000 for simple structural motor brackets because small runs interrupt mass production lines.
- **Crippling Lead Times:** Prototyping queues take 10–14 weeks, causing missed test deadlines and competition dropouts.
- **Counterfeit Silicon & Hand-Soldering Failures:** Student CubeSat flight computers frequently fail due to unverified semiconductor traders and cold joints.

1.2 The SEDS Arbitrage Solution

Through on-ground engineering coordination in Shanghai by SJTU fellows, SEDS provides:

- Direct factory pricing cutting prototype costs by **50% to 70%**.
- Rapid spindle time delivering machined components in **10 to 14 days** + air express delivery.
- Turnkey IATF 16949-certified SMT assembly with automated X-Ray BGA void inspection.
- Physical First-Article Inspection (FAI) using calibrated Zeiss Coordinate Measuring Machines (CMM) before international dispatch.

2 Global Technical Requirements & Design System

The web portal must be stack-agnostic, responsive, and seamlessly embeddable into the official SEDS Pakistan website.

Category	Specification	Technical Requirement
Target URL	/sourcing-bridge	Canonical route on sedspakistan.org
Framework	Stack-Agnostic	Next.js, React, Vue, WordPress Template, or Webflow
Color Tokens	Deep Space Navy	Background: #0B0F19, Surface Card: #111827
	Electric Blue (Primary)	Action/CTA: #2563EB, Hover: #1D4ED8
	Aerospace Cyan	Highlights: #06B6D4, Badges: rgba(6,182,212,0.1)
Performance	Emerald Success	Verified Badges: #10B981
	Mobile & Desktop	Google PageSpeed ≥ 90 ; lazy-load factory video reels

Table 1: System Design Tokens & Platform Constraints

3 Page Layout & Component Specifications

3.1 Component 1: Top Notification Bar & Header

- **Announcement Banner:** [LIVE ON-GROUND OPERATION: Active in Shanghai & GBA manufacturing clusters. 48-Hour DFM & Sourcing Benchmark open for university teams.]
- **Navigation Anchors:** #capabilities, #factory-showcase, #how-it-works, #intake-form.

3.2 Component 2: Hero Section (High-Conversion Above-the-Fold)

- **Eyebrow Tag:** SEDS SOURCING BRIDGE | ACADEMIC HARDWARE PIPELINE
- **H1 Headline:** *Direct Factory-Floor Access for Collegiate Aerospace & Hardware Teams*
- **Sub-headline:** Eliminate 12-week machine shop delays and 500% low-volume markups. Access verified 5-axis CNC machining, rapid tooling, and turnkey IATF 16949 electronics prototyping through on-ground engineering coordination in Shanghai.
- **4 Key Metric Badges:**
 1. **5-Axis Precision:** ± 0.005 mm (Al 7075-T651 / Ti-6Al-4V)
 2. **Turnkey SMT & X-Ray:** IATF 16949 (01005 passives & 0.2mm BGA pitch)
 3. **Turnaround:** 10–14 Days Spindle Time + Air Express Delivery
 4. **Quality Shield:** On-Ground First-Article CMM Inspection Included
- **Primary Call-to-Action:** [Request 48-Hour DFM & Benchmark Quote] (Smooth scrolls to #intake-form)

3.3 Component 3: Problem vs. SEDS Arbitrage Comparison Grid

3.4 Component 4: Verified Factory Floor & Machine Showcase (Gallery)

The page includes an interactive card showcase with tabbed filtering ([All], [5-Axis CNC], [Turnkey PCBA], [Tooling & GSE]):

1. **5-Axis Precision Machining Base (Kunshan/Suzhou/Shanghai):**
 - *Machinery:* DMG MORI, Haas, and Beijing Jingdiao 5-axis centers; Swiss precision lathes.
 - *Tolerances:* Linear accuracy ± 0.005 mm, hole fits H7/g6, surface finish $Ra\ 0.8\ \mu\text{m}$.
 - *Materials:* Al 7075-T651, Al 6061-T6, Titanium Ti-6Al-4V Grade 5, 316L Stainless Steel, PEEK.

Friction Dimension	Traditional Sourcing Bottleneck	SEDS Sourcing Bridge Resolution
Prototyping Cost	\$3,000–\$5,000 for simple structural brackets due to low-volume penalties.	Factory-direct rates in Kunshan/Suzhou cut costs by 50% to 70% .
Turnaround Time	10 to 14 weeks local queue time, missing competition deadlines.	10 to 14 days machining turnaround + 3 to 5 days DHL/FedEx air delivery.
Trust & Fraud	High risk of blind online scams, unresponsive brokers, and out-of-spec parts.	On-ground engineering oversight in Shanghai with Zeiss CMM verification reports .
Electronics QC	Counterfeit broker chips, hand-soldered bridges, and cold solder joints.	IATF 16949 Turnkey EMS with authorized supply chains and 100% X-Ray BGA checks.

Table 2: Friction Resolution Grid for Component 3

- *Components*: Rocket motor gimbal mounts, CubeSat bulkheads, sensor housings, propellant valves.
2. **Turnkey Aerospace Electronics Facility (Shenzhen & Zhuhai Bases):**
- *Machinery*: Europlacer prototyping line (± 0.02 mm mounting accuracy, 01005 passives, 0.2mm BGA pitch), JUKI high-speed lines, 10-zone nitrogen reflow.
 - *Testing*: Sinic-Tek 3D SPI, AI-AOI, **100% X-Ray BGA Void Inspection**, conformal coating, aging chambers.
 - *Components*: CubeSat OBCs, telemetry transceivers, Electronic Power Systems (EPS).
3. **Rapid Tooling, Molds & Ground Support Equipment (GSE):**
- *Machinery*: High-speed wire EDM, CNC gantry milling, composite layup fixtures.
 - *Components*: Carbon fiber airframe molds, static test stand plates, CNC transport foam.

3.5 Component 5: The 4-Step Sourcing Workflow

1. **Step 1 (Submit CAD)**: Upload 3D models (.STEP), 2D drawings (.PDF), or BOMs (.XLSX).
2. **Step 2 (48-Hour DFM)**: Engineering review of toolpaths, wall thicknesses, and factory benchmark quotes.
3. **Step 3 (Precision Machining)**: Spindle time on verified 5-axis machines or IATF 16949 SMT lines.
4. **Step 4 (CMM Quality Shield)**: First-Article CMM inspection in Shanghai and DHL/FedEx air delivery.

4 Engineering Intake Form Specification (#intake-form)

The conversion engine is a 12-field engineering intake form:

#	Field Name	Element Type	Req.	Validation / Placeholder
1	Full Name	input[text]	Yes	Min 2 chars; "Alex Henderson"
2	University / Lab	input[text]	Yes	"SEDS MIT / Imperial College Rocketry"
3	Institutional Email	input[email]	Yes	Valid email domain; "alex@mit.edu"
4	WhatsApp / Phone	input[tel]	Yes	International format with country code
5	SEDS Affiliation	select	Yes	Options: SEDS Pakistan, USA, UK, Canada, Other, Lab, Startup
6	Hardware Category	select	Yes	5-Axis CNC, Turnkey PCBA, Tooling/Molds, GSE
7	Material & Finish	input[text]	No	"Al 7075-T651, Hard Anodize Type III"
8	Required Quantity	input[text]	Yes	"2 prototypes + 15 flight units"
9	Target Deadline	input[date]	No	Competition or hot-fire test date
10	Critical Tolerances	textarea	No	Hole tolerances, fits, impedance, testing
11	File Upload Zone	Dropzone	Yes	.STEP, .IGES, .PDF, .ZIP up to 100MB
12	Mutual NDA Agreement	checkbox	Yes	Legally binding mutual IP protection

Table 3: Form Field Definitions & Validation Parameters

5 Cloud File Upload & Webhook Architecture

5.1 Handling 10MB–100MB Engineering CAD Files

Standard PHP and serverless functions fail on 50MB+ CAD uploads. The developer must implement **Direct Pre-Signed Cloud Storage**:

1. The client initiates upload: GET /api/get-upload-url?filename=part.step.
2. The backend issues a pre-signed PUT URL to Cloudflare R2, AWS S3, or Supabase Storage.
3. The browser uploads directly to the cloud bucket with a real-time progress bar.
4. Upon completion, the secure download URI is attached to the form payload.

5.2 Automated Webhook Notification Schema

Upon successful submission, the backend dispatches a JSON webhook to Telegram/CRM:

```
{
  "event": "sourcing_inquiry_received",
  "timestamp": "2026-08-31T23:10:00Z",
  "lead": {
    "name": "Alex Henderson",
    "university": "Imperial College London Rocketry",
    "email": "alex@imperial.ac.uk",
    "phone": "+44 7700 900077",
    "chapter": "UKSEDS",
    "category": "5-Axis Precision CNC Machining",
    "material": "Al 7075-T651 Hard Anodized",
    "quantity": "2 prototypes + 15 flight units",
    "file_download_url":
      "https://storage.sedspakistan.org/cad-uploads/20260831_imperial_gimbal.step"
  }
}
```

6 Developer Verification Checklist

- ☐ Fully responsive layout verified on Mobile (iOS Safari / Android Chrome) and Desktop.
- ☐ Dropzone successfully uploads 3D .STEP, .PDF, and .ZIP files up to 100MB.
- ☐ Instant webhook firing delivers lead contact and download link to Telegram/Email.
- ☐ Client-side validation active on all required fields with high-contrast error states.
- ☐ Initial page load speed achieves Google PageSpeed score ≥ 90 .